

Brian Chu

SOFTWARE DEVELOPER · CHEMICAL ENGINEER

☎ (415)-969-0765 | ✉ brian.chu1030@gmail.com | 🏠 bchu-ops.github.io/pumpkio9-website/ | 💻 bchu-ops | 🌐 brian-chu123

Summary

Masters in Chemical Engineering in University of California, San Diego and recent Computer Science graduate and specialty in ML frameworks. Proven experience as a AI/ML Engineer at Vulcan Engineering, contributing to a 45% improvement in R-CNN model accuracy. Experienced in developing scalable, production-level software and collaborating with cross-functional teams.

Work Experience

Vulcan Engineering Solutions

Dec. 2024 - Present

SOFTWARE ENGINEER INTERN

- Implemented and researched **scalable R-CNN models** and used OpenCV for image analysis, improving defect detection by 45%.
 - Collaborated with full-stack teams to pipeline and preprocess data, enhancing model readiness and customer experience.
 - Developed a **LangChain-Groq LLM chatbot** that helps structural engineers interpret design documents and standards (e.g., ASCE 7).
- Skills: Python, Microsoft Azure, LangChain, Hugging Face Transformers, PyTorch

Sustainable Power and Energy Center

Jun. 2022 - Aug. 2022

LAB INTERN

- Planned and executed independent battery performance tests of coin-cell batteries, **increasing reliability insights by 50%**.
- Applied research from seminars to revise chemical formulas, **enhancing fabrication efficiency of battery by 15%**.

Senior Design Project Team Member (UCSD)

Jan. 2023 - Jun. 2023

PROCESS ENGINEER

- Designed a Methanol to Dimethylamine plant with ASPEN, **optimizing conversion rate by 85%** while lowering electricity usage by 55%.

Skills / Technologies

Languages	Python, Java, SQL, C++, JavaScript
AI/ML Frameworks	SpaCy, Keras, TensorFlow, LangChain, Hugging Face, Seaborn, Scikit-Learn, RMI, Django, React, AWS Lambda
Engineering Focus	Energy Systems, Manufacturing Automation, Process Modeling, Data-Driven Simulation
Simulation & Tools	MATLAB, AutoCAD, SolidWorks, ASPEN, COMSOL, BTDSA, Arduino
Certifications & WIP	FE Chemical, AutoCAD Certified Associate
Interests	Soccer, Formula 1, Cars, ML/AI, Control Systems, Battery Technology, Plasma Physics

Projects

Natural Language Processing (NLP) Drug Drug Interactions

San Diego, CA

SKILLS: Python, spaCy, Seaborn, Hugging Face Transformers, PyTorch

- Implemented KGNN from similar research using Tensorflow and Keras
- Performed tokenization of drug effects while generating corpus for each description of combined drug effects

Provinces Enterprise System

San Francisco, CA

SKILLS: Java, MySQL, HTML, CSS

- Loaded 500 providence-capital matches from a MySQL database to implement a multi-threading client-server system
- Implemented Creational Design Patterns such as Abstract Factory Classes, Builder Classes, Prototyping

Personal Website

San Diego, CA

SKILLS: Javascript, React, Python, TensorFlow

- Built full-stack web apps using React, Python, and TensorFlow; integrated APIs (Spotify, Google Maps) for live data visualization and authentication with Nailong cartoon videos
- Applied responsive design and state management best practices to improve UX for dynamic, data driven applications

Speed Demon

Taipei, Taiwan

SKILLS: SolidWorks, Python, PostgreSQL, OpenAI, Chroma Vector DB

- Intelligent suggestion of throttle inputs and steering to improve lap-times with persistent inventory memory using PostgreSQL and OpenAI
- Constructed chassis of a go-kart on SolidWorks aligning with FSAE regulations

Education

University of California, San Diego

June 2025

MS, CHEMICAL ENGINEERING

University of California, San Diego

June 2024

BS, COMPUTER SCIENCE W/ SPEC IN ML/AI AND CHEMICAL ENGINEERING